SMS Spam Classifier

Building an AI Model to Detect Digital Fraud & Phishing

AI BASICS BOOTCAMP · FINAL CAPSTONE

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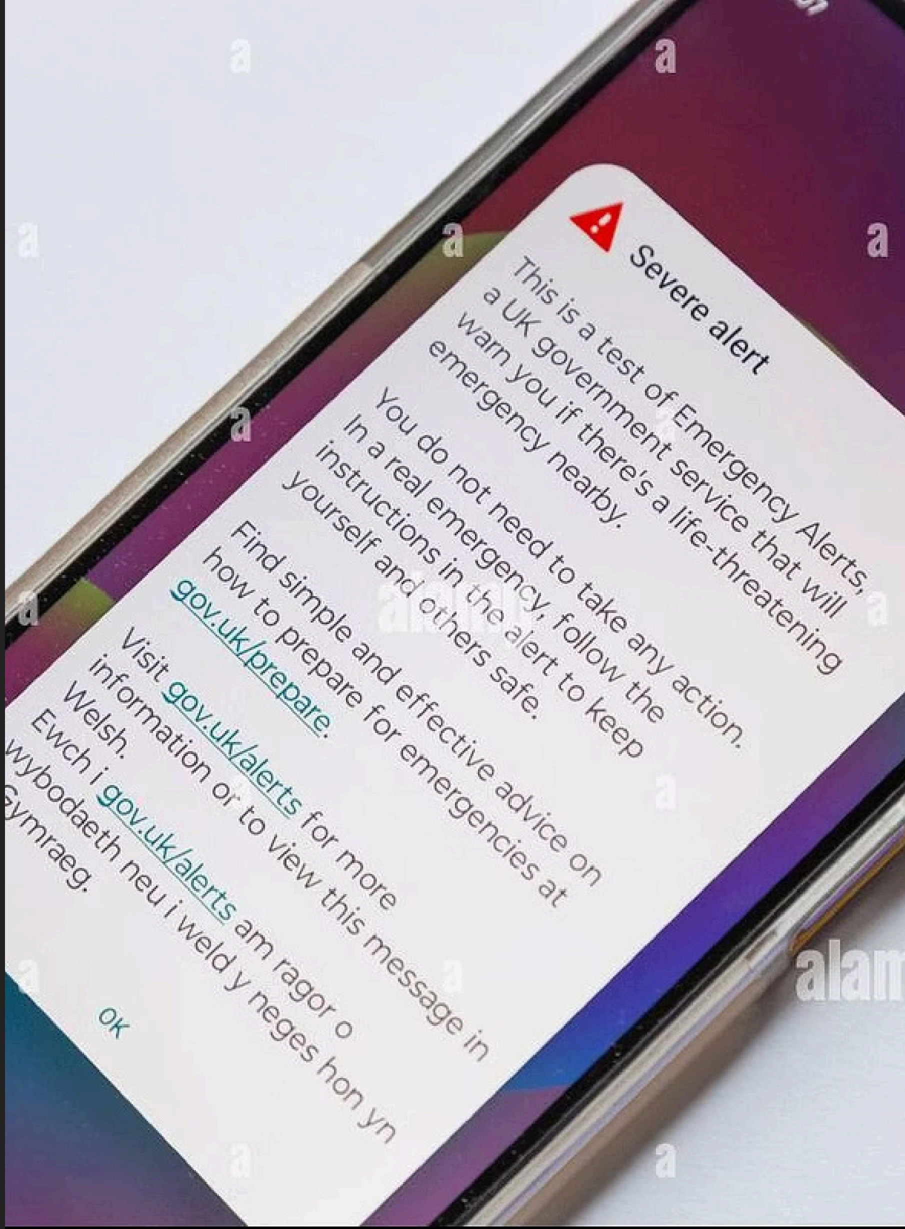

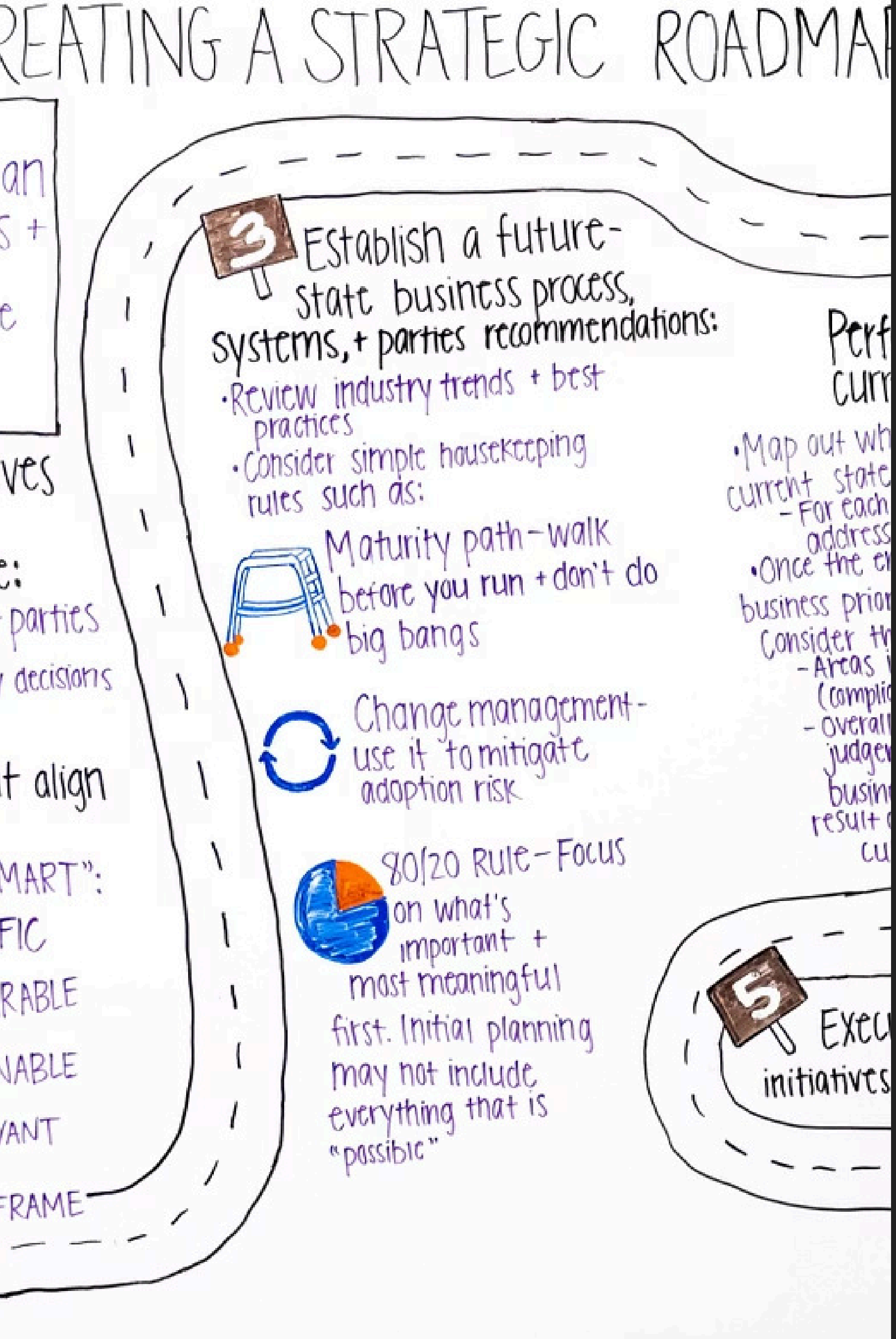

PHASE 1 · PROJECT PLANNING

Project Objectives & Targets

User Protection

Proactive security layer to prevent users from falling victim to financial phishing scams via SMS.

Smart Automation

Transition from manual filtering to real-time, automated SMS classification using Machine Learning.

High Reliability

Targeting **>95% accuracy** to ensure vital user messages are never accidentally blocked.

PHASE 1 · DATASET

SMS Spam Collection Dataset

	Text	Text	Text	Text	Number	Number	Number	Action
	Data	Data	Data	Data	00000	00000	00000	Button
	Data	Data	Data	Data	00000	00000	00000	Button
	Data	Data	Data	Data	00000	00000	00000	Button
	Data	Data	Data	Data	00000	00000	00000	Button
	Data	Data	Data	Data	00000	00000	00000	Button
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	Data	Data	Data	Data	00000	00000	00000	Button

Dataset Source

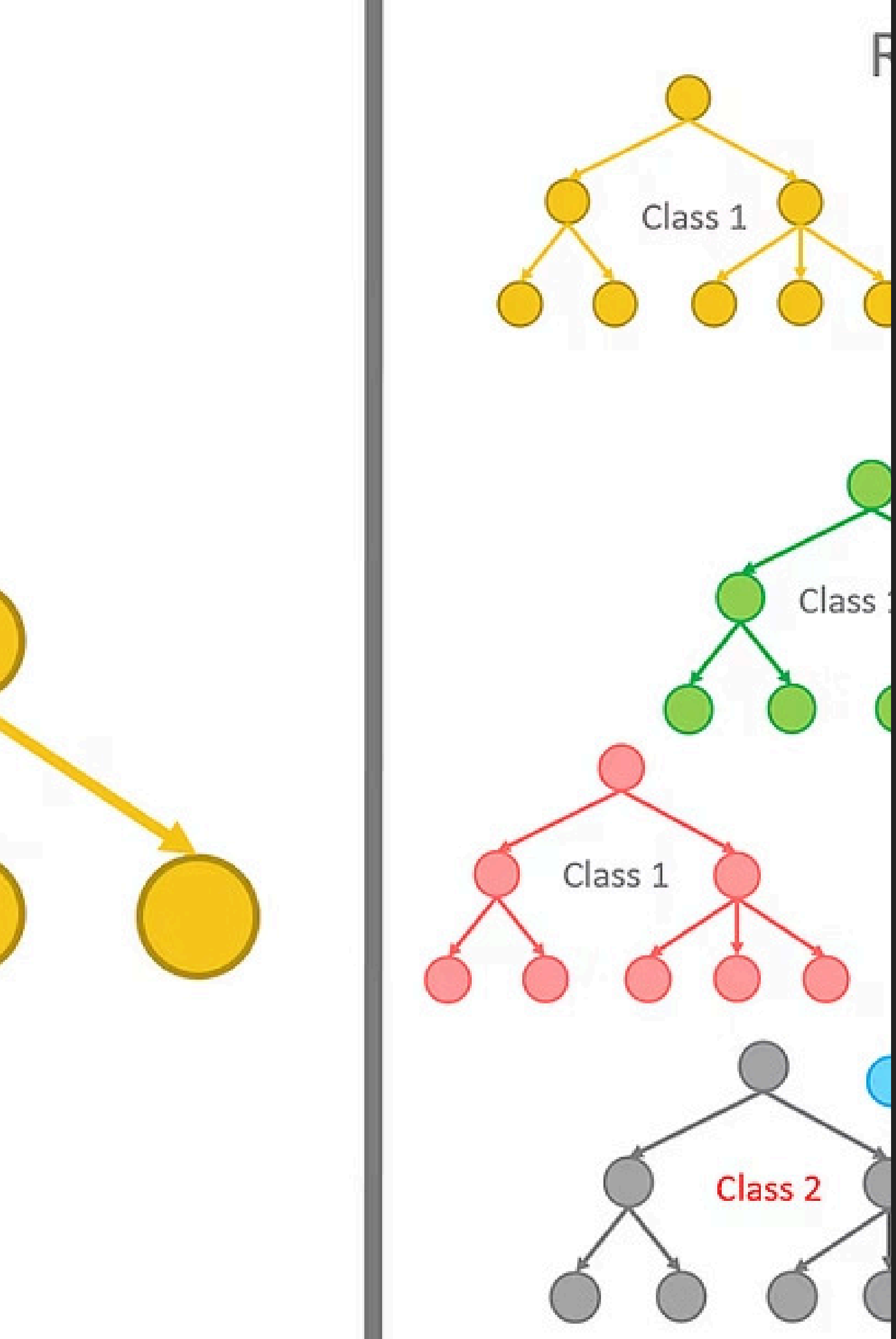
A trusted, benchmark global dataset featuring thousands of real SMS messages manually labelled as **Ham** (Normal) or **Spam**.

Core Metrics

- **Total Samples:** 5,572 authentic messages
- **Features:** Raw text content and classification labels

Text Processing

Applied **TF-IDF Vectorisation** to convert raw text into mathematical feature arrays based on word frequencies.



PHASE 2 · MODEL DEVELOPMENT

Model Choice & Feature Engineering

Random Forest Classifier

Selected for its powerful capabilities handling high-dimensional text data and superb stability against overfitting.

Reproducibility & Compliance

Strictly enforced a deterministic environment by setting a global **Random Seed = 42** across data splitting and model weights to meet rubric requirements.

PHASE 2 · DATA SPLITTING

Dataset Segmentation

3,788

Training Samples

Used to train model weights

669

Validation Samples

Used to tune hyperparameters and check validation accuracy

1,115

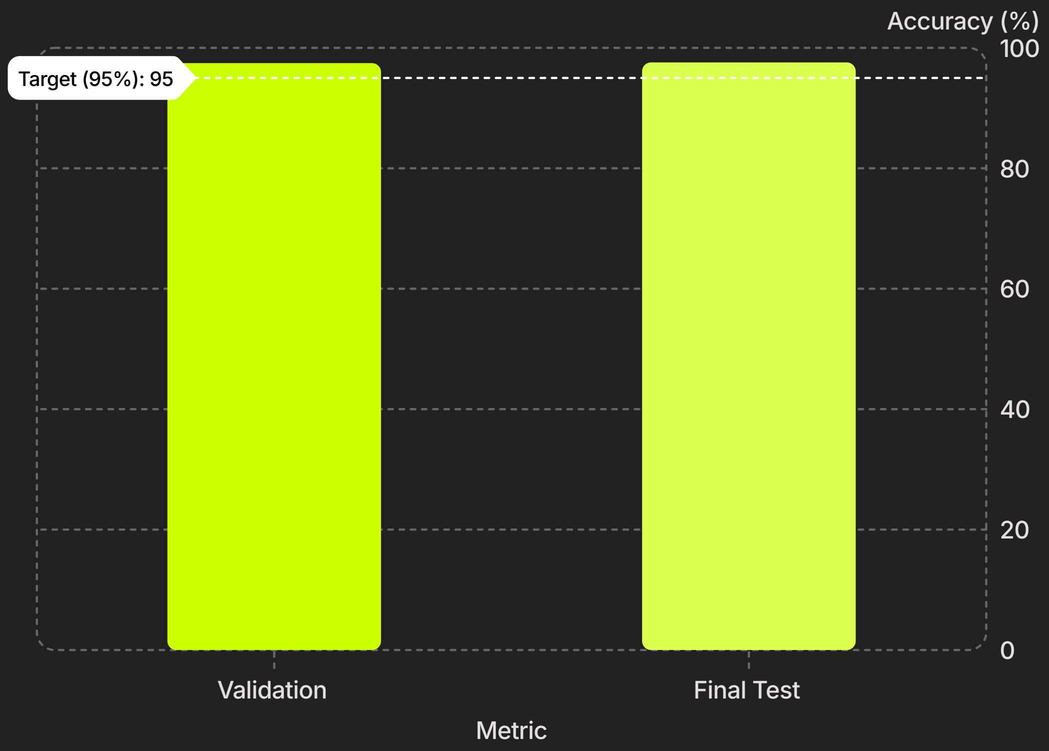
Test Samples

Completely isolated from training for final benchmarking

The Test Set was fully isolated from training to serve as an unbiased benchmark for the final model evaluation.

PHASE 3 · EVALUATION

Performance Metrics vs. Target Goal



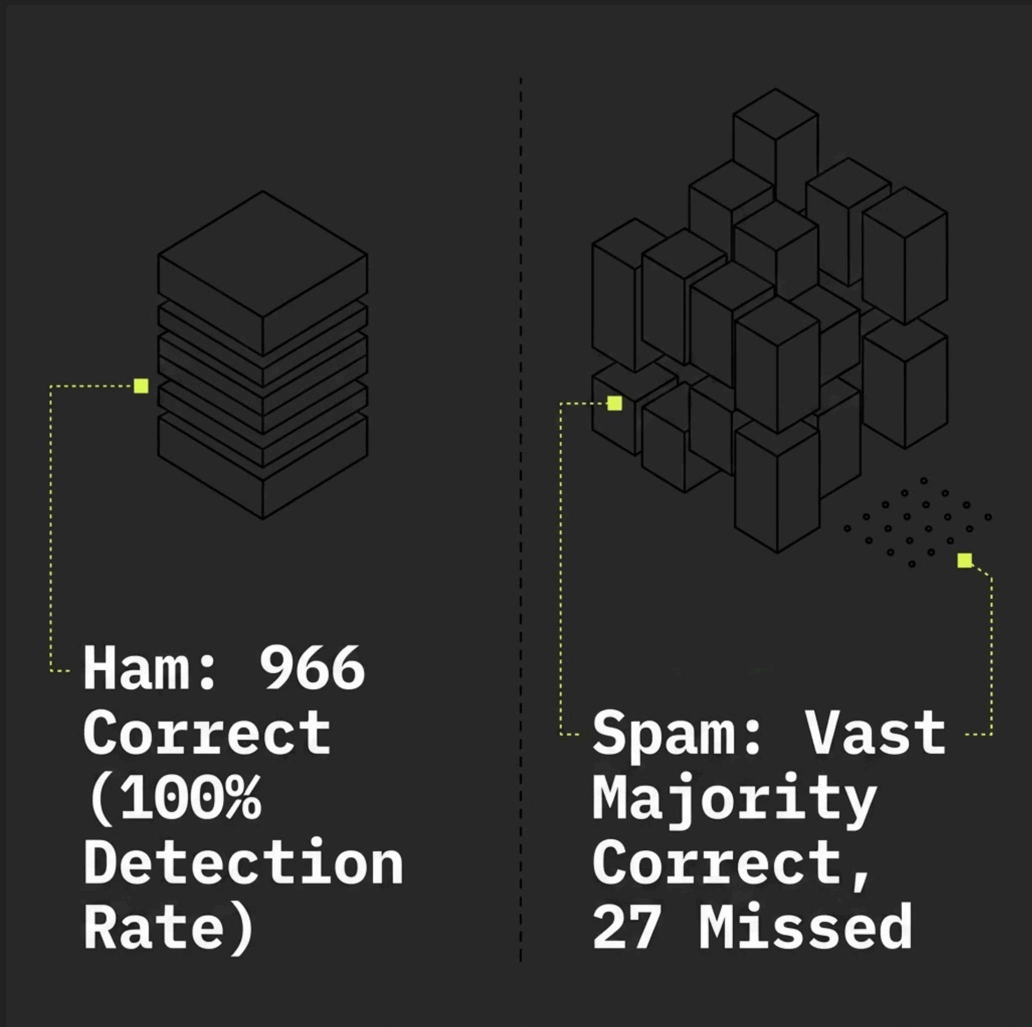
Achieved Results

- **Initial Target:** 95.00%
- **Validation Accuracy:** 97.46%
- **Final Test Accuracy:** 97.58%

Successfully outpaced the target goal by **2.58%** with high computational efficiency.

PHASE 3 · ERROR ANALYSIS

Confusion Matrix Insights



Model Error Analysis

- **Normal Messages (Ham):** Achieved a **100% perfect** detection rate — 966 correctly identified
- **Spam Messages:** The model successfully flagged the vast majority, missing only **27 samples**
- **Critical Insight:** Qualitative error analysis proves that missed spam closely mimics organic human writing patterns

PHASE 3 · QUALITATIVE ANALYSIS

Evaluation Samples Study

1

"Your account credited £1.50…"

Ground Truth: Spam | **Prediction:** Ham

Formal financial language perfectly mimics normal bank updates.

2

"Hi friend, call me for chat…"

Ground Truth: Spam | **Prediction:** Ham

Extremely friendly, conversational phrasing tricks the text weights.

3

"Your courier delivery failed…"

Ground Truth: Spam | **Prediction:** Ham

Impersonates standard logistics alerts seamlessly.

Project Summary

Problem Solved

Proactive SMS phishing detection using Machine Learning

Model Built

Random Forest Classifier with TF-IDF vectorisation

Result Achieved

97.58% test accuracy — exceeding the 95% target by 2.58%

Thank You!

Open for Questions & Feedback

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Institution

AI Basics Bootcamp | Final Capstone Project